HARSHITA RATHORE

Python Backend Developer | FastAPI | Generative AI & RAG

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PROFESSIONAL SUMMARY

Python backend developer focused on building API-driven applications with FastAPI, SQL, and PostgreSQL. Hands-on project experience integrating Generative AI, LLM APIs, Retrieval-Augmented Generation (RAG), and agentic workflows. Familiar with vector search using FAISS, database operations with SQLAlchemy, JWT-based authentication, and containerization fundamentals.

EDUCATION

Medi-Caps University, Indore — B.Tech, Computer Science Engineering | 2022–2026
CGPA: 8.26

Bal Niketan Higher Secondary School, Indore — Class XII, CBSE | 2020–2022
Class X: 94.67% | Class XII: 73.6%

KEY SKILLS

Programming & Backend: Python, FastAPI, REST APIs
GenAI & Applied AI: RAG Pipelines, LLM Integration, Agentic AI (CrewAI), Generative AI, Basic Machine Learning, Basic NLP
Data & Databases: FAISS (Vector Database), MySQL (RDBMS), PostgreSQL, SQLAlchemy
Cloud & DevOps: Docker (Basic), GitHub Actions, AWS EC2 (Basic), CI/CD Fundamentals
Tools: Git, GitHub, VS Code, Google Colab, Postman, Swagger UI

PROFESSIONAL EXPERIENCE

Generative AI & Agentic Systems Intern — Datagami (NASSCOM × IBM Initiative) | **Jan 2026 – Apr 2026**

- Built a Retail RAG Chatbot using FastAPI, FAISS vector store, and Google Gemini to answer contextual questions about a retail product catalog.
- Developed a four-agent CrewAI workflow (Planner → Researcher → Analyst → Critic) using Groq LLM and DuckDuckGo search for query decomposition and synthesis.
- Reduced Docker image size by approximately 60% by replacing Hugging Face embeddings with GoogleGenerativeAIEmbeddings, reducing build size and startup overhead.
- Containerized the application with Docker and automated image builds and deployment steps using a GitHub Actions CI/CD pipeline.

AI / ML Intern — Medi-Caps University (In-House Research Lab) | **Jun 2025 – Jul 2025**

- Gained introductory exposure to Python-based machine learning workflows, including data preparation, model training, and basic evaluation.
- Practiced foundational concepts in supervised learning and explored image-classification workflows during guided internship tasks.

PROJECTS

Retail RAG Chatbot + Agent Mode | **GitHub**
FastAPI, FAISS, Gemini, CrewAI, Groq, DuckDuckGo, Docker, GitHub Actions, AWS EC2

- Designed an end-to-end RAG pipeline with a FAISS vector store for semantic retrieval; added a modular Agent Mode toggle integrating CrewAI multi-agent workflows.
- Containerized the application with Docker and deployed it on AWS EC2; automated the build and push pipeline using GitHub Actions.

StockSphere — Intelligent Inventory Management with Generative AI | **GitHub**
Python, FastAPI, React, PostgreSQL, SQLAlchemy, JWT, Google Gemini API

- Developed a full-stack inventory management application with product CRUD operations using React, FastAPI, and PostgreSQL.
- Used SQLAlchemy for database operations and implemented JWT-based authentication with user-specific inventory retrieval.
- Integrated Gemini for AI-assisted product description generation and implemented Python-based stock classification with an inventory-advice endpoint.

CERTIFICATIONS

- SQL — HackerRank
- Python Essentials — Cisco Networking Academy
- Generative AI — AWS
- Introduction to Large Language Models — Google Cloud Skills Boost